

CHEMISTRY IX

10 Year Questions Distribution

According to the Chapters.

This Distribution is carried out According to
NEW BOOK OF CHEMISTRY.

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Section "B" (Short Answer Question)

QUESTIONS	Years
Chapter 1: FUNDAMENTALS OF CHEMISTRY	
Section B	
2 Write down about the applications of Chemistry in daily life. 3. Differentiate between Molecular and Empirical formula. 4. Complete and balance the following equations: (i) $KNO_3 \longrightarrow$ (ii) $Ca + H_2O \longrightarrow$ (iii) $NaHCO_3 \longrightarrow$ 11. Write down the chemical formulae and one use of each of the following compounds (i) Soda Ash (ii) Baking Soda (iii) Washing Soda	2025
4. The following statements belong to which branch of Chemistry? Identify: (i) That chlorofluoro compounds are responsible for depletion of Ozone layer. (ii) That investigates the properties of Ammonia gas. (iii) That find out the quantity and quality of Paracetamol in Panadol tablet. 8. Balance the following chemical equations by inspection method: (i) $CaCO_3 + HCl \rightarrow CaCl_2 + H_2O + CO_2$ (ii) $KClO_3 \rightarrow KCl + O_2$ (iii) $NH_3 + O_2 \rightarrow NO + H_2O$ 9. Calculate the number of atoms presents in 9.2 g of Calcium. (Atomic mass of Ca = 40) 11. Write down the chemical formulae of the following compounds: (i) Caustic Soda (ii) Sand (iii) Sugar	2024

(iv) Lime Stone	(v) Gypsum	(vi) Baking Soda	
3. Calculate the number of moles and number of molecules present in 45 grams of glucose ($C_6H_{12}O_6$).			2023
6. Balance the following chemical equations by inspection method:			
(i) $KNO_3 \longrightarrow KNO_2 + O_2$			
(ii) $Ca + H_2O \longrightarrow Ca(OH)_2 + H_2$			
(iii) $CO + O_2 \longrightarrow CO_2$			
9. Balance the following chemical equations:			2022
(i) $Ca + O_2 \rightarrow CaO$			
(ii) $KClO_3 \rightarrow KCl + O_2$			
(iii) $NH_3 + O_2 \rightarrow NO + H_2O$			
12. Write down the chemical formulae of given compounds:			
(i) Lime stone	(ii) Washing soda	(iii) Sugar	
2. Write three sentences on the importance of Chemistry.			2021
3. Define mole, calculate the number of moles in 40 grams of water (H_2O) (At Mass $O=16$, $H=1$).			
6. Balance the following chemical equations by using co-efficient.			
(i) $Si + Cl_2 \rightarrow SiCl_4$			
(ii) $C + O_2 \rightarrow CO_2$			
(iii) $Zn + HCl \rightarrow ZnCl_2 + H_2$			
10. Write down the basic units of the following quantities in system international:			
(i) Mass	(ii) Length	(iii) Temperature	
(iv) Electric Current	(v) Time	(vi) Quantity of substance	
2. Chemistry and Human society "Go Hand in Hand". Write four arguments in support of this statement.			2019
7. Find out the mass of 4.5 moles of the following compounds:			
(i) Water (H_2O) (ii) Sodium Carbonate (Na_2CO_3)			
Atomic masses: ($Na=23$, $C=12$, $H=1$, $O=16$)			
2. What is Chemistry? Write names of any six branches of Chemistry.			2018
7. How many atoms or molecules are there in:			
(i) 8g of Sulphur (S) (ii) 8.8g of Carbon dioxide (CO_2)			
(Atomic Masses : $S=32$, $C=12$, $O=16$)			
8. Reproduce any two of the following chemical reaction in the form of balanced chemical equations:			
(i) Carbon monoxide reacts with Oxygen gas to form Carbon dioxide			
(ii) Zinc metal reacts with Hydrochloric acid to give Zinc chloride and Hydrogen gas.			
(iii) Potassium Chlorate decomposes on heating into Potassium Chloride and Oxygen gas			
2. Write names of any six branches of Chemistry.			2017

Explain with examples.

Q. 8 Explain why hydrogen and oxygen are considered as element whereas water is compound?

Section C

Q. 1 What do you mean chemical species, explain ion, molecular ion and free radical?

Q. 2 Writ down the applications of chemistry in daily life?

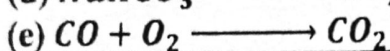
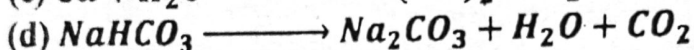
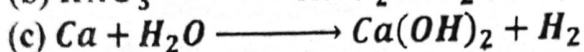
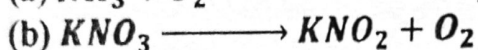
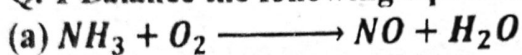
Q. 3 Explain in detail empirical and molecular formula?

Q. 4 Explain the steps for balancing the equation?

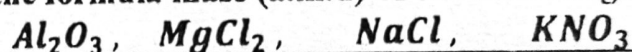
Q. 5 Name the branches of chemistry and discuss any five branches?

Section D

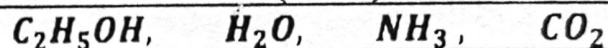
Q. 1 Balance the following equations by inspection method:



Q. 2 Calculate the formula mass (a.m.u) of the following?

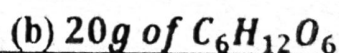


Q. 3 Calculate the molecular mass (a.m.u) of the following?



Q. 4 How many moles are required to prepare 40 gm of H_2SO_4 ?

Q. 5 Calculate the number of moles and number of molecules present in the following?



Chapter 2: ATOMIC STRUCTURE

Section B

5. Write down the electronic configuration of Na, Mg and Al While their atomic number are $\text{Na} = 11$ $\text{Mg} = 12$ and $\text{Al} = 13$ **2025**

7. What are isotope? Describe briefly three isotopes of Uranium. **2024**

4. Write down the electronic configuration of the following elements: **2023**

(i) Carbon

(ii) Fluorine

(iii) Sodium

12. Write down three properties of cathode rays.

5. What are isotopes? Describe isotopes of Chlorine.

7. Write down any three points of atomic model proposed by Schrodinger. **2022**

4. Write down the electronic configuration or arrangements of the following Elements according to the Bohr's Atomic Model.

2021

Elements	Atomic Numbers	K 1 st shell	L 2 nd shell	M 3 rd shell	N 4 th shell
Hydrogen	01				
Nitrogen	07				
Neon	10				

9. Define the following: (i) Isotopes

2018

5. How many protons and neutrons are present in the following atoms: (i) ${}^{14}_7\text{N}$ (ii) ${}^{27}_{13}\text{Al}$ (iii) ${}^{23}_{11}\text{Na}$

2017

5. Write down three characteristics of Cathode Rays.

2016

8. Define Isotopes and find out the number of neutrons in the following Isotopes. (i) ${}^{207}_{82}\text{Pb}$ (ii) ${}^{234}_{92}\text{U}$

2015

17. Chlorine is naturally found in two stable isotopes ${}^{35}_{17}\text{Cl}$ and ${}^{37}_{17}\text{Cl}$ in the proportion of 75% and 25% respectively. Calculate its atomic mass.

Section C

14. How did Rutherford present the atomic model? Describe his experiment in detail and explain the atomic model presented by him.

2025

17. Write down six postulates of Bohr's Atomic Model.

2024

16. An atom has only one electron in its 'M' shell then what would be its atomic number? Identify that element and write four common characteristics of the group from - which it belongs to.

2023

15. (OR) Describe discharge tube experiment with labelled diagram and state what were the justification and conclusion of J.J. Thomson in this respect

2022

SOME IMPORTANT QUESTIONS FROM NEW BOOK

Section B

Q. 1 Draw the structure of isotopes of chlorine to justify the definition of isotopes?

Q. 2 An atom has 5 electrons in M shell than:

(a) Find out its atomic number?

(b) Write Electronic configuration of atom?

(c) Name the element of atom?

Q.3 Justify that Rutherford atomic model has defects?

Q. 4. Describe wave particle duality of electron of De Broglie Hypothesis?

Q. 5. What are Limitations of Bohr's Atomic Model?

Q. 6. Differentiate between shell and sub shell with examples?

Q. 7. How the atoms of O_8^{17} and O_8^{18} are similar or different from each other?

Q. 8. Write down the names of sub atomic particles their masses in a.m.u with their unit charges.

Section C

Q. 1. Discuss Rutherford's gold metal foil experiment in the light of structure of atom.

Q. 2. Write down the applications of isotopes in daily life.

Q. 3. Explain how Bohr's atomic model is different from Rutherford atomic model.

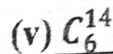
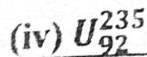
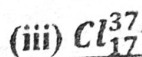
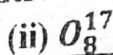
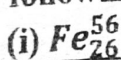
Q. 4. Prove that modern theory of De Broglie is related with Einstein and Planck's equations.

Q. 5. How are cathode rays produced? What are their major characteristics?

Q. 6. Describe the Schrodinger atomic model.

Q. 7. Describe briefly the experiments which provide clue and evidences of electron, proton and neutron in an atom.

Q. 8. How many protons, neutrons and electrons are present in the following elements?



Chapter 3: PERIODIC TABLE AND PERIODICITY OF PROPERTIES

Section B

7. What is meant by an electronegativity? Describe its trend in group and period with examples. **2025**

3. Write down three differences between Alkali metals and Alkaline earth metals. **2024**

6. What is atomic radius? Discuss its trend across the period and group in the periodic table.

2. What is meant by the terms 'period' and 'group' with respect to periodic table? Identify period and group of 7N and ${}^{11}Na$ **2023**

3. Write down three differences between Alkali metals and Alkaline earth metals. **2022**

6. Identify group and period in the periodic table of the following elements: (i) ${}_3Li$ (ii) ${}_6C$ (iii) ${}_{13}Al$

6. Write down four salient features of Mendeleev's periodic table. **2018**

9. Define the following:

(ii)Electronegativity	(iii) Electron Affinity	
6. Define:	(i)Dobereiner's rule of Triad	
(ii) Newland's law of octave	(iii) Modern Periodic Law	2017
6. State the following laws:	(i) Dobereiner's Law of Triad	
(ii) Newland's Law of Octave	(iii) Modern periodic Law	2016
12. State following laws: (i)Modern Periodic law		
21.What are metalloids? Write two differences between metals and non-metals.		2015

Section C

18. How elements are arranged in periodic table? Describe groups and periods in periodic table.	2025
16. Identify the highly electronegative group of Periodic table. Enlist the elements present in this group. Give three common characteristics of these elements.	2024
12. Write the name of the elements of group "I-A" of the periodic table. What does the group I-A elements called in the periodic table Write "Four" general properties of this group.	2021
18. (b) Write the salient features of Group VIIIA (Inert or Noble gases) and describe the discovery of Noble gases	2019
25(b). Write four general properties of transition elements.	2017
24 (c). What are elements of Group I-A of Periodic Table called? Write three general properties of these elements.	2016
24 (b). Name the group of the periodic table which contains highly electronegative elements . Also write three common characteristics of this group	2015

SOME IMPORTANT QUESTIONS FROM NEW BOOK

Section B

Q. 1. Distinguish between periods and groups.
Q. 2. Describe the trend of electronegativity within group and period with the help of examples?
Q. 3. Explain the similarity of chemical and physical properties of elements in the same family.
Q. 4. Justify that periodicity of properties dependent upon number of protons in an atom?
Q. 5. Identify that which halogens exist as gases, liquid and solid?
Q. 6. Why Alkaline earth metals shows irregular melting and boiling point?
Q. 7. Why ionization energy, electron affinity and electronegativity exhibit same trend in period and groups?

Section C

Q. 1. Discuss in detail the long form of periodic table.

Q. 2. Determine the demarcation of periodic table in to s, p, d and f blocks.

Q. 3. Identify the electronic configuration of the following elements.

Na, Ca, F, Si

Q. 4. Determine the location of families on periodic table.

Q. 5. Discuss that Mendeleev periodic law provide a base for modern periodic table.

Q. 6. Explain how shielding effect influence the periodic trends?

Chapter 4: CHEMICAL BONDING

Section B

8. What is meant by lone pair and unpaired electrons? How many lone pair and unpaired electrons are there in Nitrogen atom? **2025**

10. Draw electronic dot and cross structures of following molecules: (i) H_2 (ii) N_2 (iii) O_2 **2024**
(Atomic number of H = 1, N = 7, O = 8)

10. Write down three important characteristics of covalent compounds. **2023**

5. Write down three comparative points between Covalent Bond and Dative Bond.(at least 3 points each) **2021**

5. Write down four comparative points between Covalent bond and Dative bond. **2019**

9. Define the following: (iv) Polar Bond **2018**

7. Write three characteristic properties of Ionic compounds. **2017**

7. Write three differences between ionic compounds and covalent compounds **2016**

10. Define:

(i) Chemical Bond (ii) Metallic Bond (iii) Hydrogen Bond

Section C

15. Why do some molecules form covalent bond? Write its three types with examples. **2025**

18. What is an ionic bond? Explain the formation of ionic bond between Sodium and Chlorine ionic compound Sodium chloride is formed. **2024**

15. Explain the importance of glues and epoxy resins in our society. **2023**

17. What is coordinate covalent bond? Explain it by the formation of Ammonium Ion (NH_4^+) and hydronium ion (H_3O^+)

15. Define ionic bond. Illustrate the bond formation in Sodium Chloride (NaCl) also write any two characteristics of ionic compounds.

2022

23(b). (c) Write down 4 Characteristics of Covalent Compounds.

24 (c). Define: (i) Chemical Bond

(ii) Ionic Bond

(iii) Covalent bond

(iv) Co-ordinate covalent bond.

2015

SOME IMPORTANT QUESTIONS FROM NEW BOOK

Section B

Q. 1. Draw dot and cross diagrams to show how different types of chemical bonds are formed when fluorine reacts with:

(a) Hydrogen

(b) potassium

Q. 2. What is meant by octet and duplet rule?

Q. 3. Can you draw an ion which is formed by the atom losing three electrons?

Q. 4. How oxygen forms an anion?

Q. 5. What is the differentiate between lone pair and bond pair?

Q. 6. Explain why table salt has a very high melting point.

Q. 7. How is electronegative value, determined the formation of chemical bond?

Q. 8. Why is too easy for magnesium atom to lose two electrons?

Q. 9. Atoms of metallic elements can form ionic bond, but they are not very good to form covalent bonds. Why?

Q. 10. How does an ion differ from an atom?

Q. 11. Describe dipole-dipole forces.

Q. 12. Write uses of adhesive material.

Q. 13. Why Intermolecular forces are weaker than intra-molecular forces?

Q. 14. Write characteristics of metallic bond.

Q. 15. Covalent bonds are strong and hard to break but why most of the covalent compounds have low melting and boiling points.

Q. 16. Write down the characteristics of ionic compounds.

Q. 17. Why ionic compounds are solid?

Q. 18. How is hydrogen bonding affecting the physical properties of compounds?

Q. 19. Complete the chart:

Atomic Number	Number of Protons	Number of Electrons	Number of Neutrons	Electronic configuration	Number of valence electrons

11	11	11	12	2, 8, 1	1
12					
13					
14					
15					
16					

Section C

Q. 1. Define ionic bond. Discuss the formation of sodium chloride (NaCl).

Q. 2. Explain element attain stability?

Q. 3. Describe the formation of a covalent bond between two nonmetallic atoms. Explain single, double and triple covalent bond with examples.

Q. 4. How are electrons arranged in molecular compound? Draw electron dot and cross structures for the following atoms.

(a) H_2O (b) N_2 (c) CH_4 (d) C_2H_2 (e) Cl_2 (f) H_2

Q. 5. Define metallic bond. How metallic bonds are formed?

Q. 6. What is coordinate covalent bond? Explain with two examples.

Q. 7. What do you understand about ionic character of covalent bond?

Q. 8. Differentiate the properties of polar and non-polar compounds.

Q. 9. Explain the importance of glues and epoxy resins in our society.

Chapter 5: PHYSICAL STATES OF MATTER

Section B

6. The pressure of a sample gas is 8 atm and the volume is 15 liters. If the pressure is reduced to 6 atm, what will be its new volume? **2025**

7. The pressure of a sample gas is 3 atm and the volume is 5 litre. If the pressure is reduced to 2 atm then what would be its new volume? **2023**

8. Write down three differences between amorphous solids and crystalline solids.

11. 60ml sample of a gas is heated from 27°C to 52°C , at constant pressure. What will be final volume of this gas? **2022**

16. Give reason:

(i) Which gas among CO_2 , CH_4 and H_2 will diffuse faster and why?

(ii) Water filled glass bottles often crack in freezer. Why

8. Define: (i) Fusion (ii) Evaporation **2018**

8. Define: (i) Fusion (ii) Evaporation **2017**

11. Define: (i) Diffusion (iii) Boiling Point **2016**

Section C

pressure. What is the final volume?

Q. 4. A gas occupies a volume of 40.0 dm^3 at standard temperature (0°C) and pressure (1 atm), when pressure is increased up to 3 atm unchanged temperature what would be the new volume?

Q. 5. The 800 cm^3 of a gas is enclosed in a container under a pressure of 750 mm of Hg. If the volume is reduced to 250 cm^3 , what will be the pressure?

Chapter 6:

SOLUTION

Section B

10. Calculate the percentage concentration (m/m) of the solution by dissolving 15g salt in 110g water. **2025**

5. How do you prepare 100ml of 0.4M MgSO_4 from a stock solution of 2.0M MgSO_4 ? **2024**

12. What is meant by Tyndall effect? How is it used for the identification of a true solution and a colloidal?

9. Define the following:

(i) Saturated solution

(ii) Unsaturated solution

(iii) Super saturated solution

11. How much sodium hydroxide (NaOH) is required to prepare 400 cm^3 of 0.3 M solution? **2023**

4. What is an aqueous solution? Give an example. Air is a mixture of gases then why Nitrogen gas is considered solvent? **2022**

8. What is the mass of Oxalic acid ($\text{C}_2\text{H}_2\text{O}_4$) required to prepare 100 cm^3 volume of 2M Oxalic acid solution? (where Atomic masses: $\text{C} = 12$, $\text{H} = 1$, $\text{O} = 16$)

10. —OR— Mention any three characteristics of solution. **2021**

12. Calculate molality of the aqueous solution containing 18 g Glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) dissolved in 300g of water. **2019**

19.(a) Define solubility and describe in detail the factors that affect solubility. **2018**

19. What is suspension? Write names of four examples of suspension in daily life. **2017**

21. Calculate the molarity of a solution that contains 2 moles of NaOH dissolved in 0.5 liter of the solution.

9. Define solubility and describe effects of temperature and pressure on solubility. **2016**

13. Write three differences between Solution and Suspension.

11. Define: (i) Solubility (ii) Molarity (iii) Molality **2015**

Section C

17. Compare properties of Solution, Suspension and Colloids. (Write eight points of each)	2025
18. Compare the characteristics of solution, colloid and suspension,	2023
13(ii). Calculate the molarity of a solution containing 4 grams of sodium Hydroxide (NaOH) in 250 cm ³ of solution.	2021
25(b). Write four difference between solution and suspension.	2016
SOME IMPORTANT QUESTIONS FROM NEW BOOK	

Section B

- Q. 1. Explain the solute-solvent interaction to prepare sodium chloride solution.
- Q. 2. Differentiate between the saturated and unsaturated solution?
- Q. 3. Define solution and explain the major components of solution?
- Q. 4. What do you mean by mass/volume percent?
- Q. 5. Define with example one molar solution.
- Q. 6. Why does colloidal show Tyndall effect?
- Q. 7. Define the terms
i. Dilution ii. Concentration iii. Solubility iv. Molarity
- Q. 8. Polar and ionic solutes dissolve in polar solvent only. Why?
- Q. 9. Why does Polar solute not dissolve in nonpolar solvent?
- Q. 10. How are solutions beneficial for community?
- Q. 11. Why salt dissolves in water?
- Q. 12. Air is a solution containing oxygen, carbon dioxide, nitrogen and other gases. Which one of the gas is called solvent and why?
- Q. 13. Why gasoline does not dissolve in water?

Section C

- Q. 1. Describe how to prepare dilute solution from concentrated solution.
- Q. 2. Define the term solubility. How does nature of solute and solvent determine the extent of dissolution?
- Q. 3. Why the solubility of a salt increases with the increase in temperature?
- Q. 4. Explain the attraction of Na^+ and Cl^- ions for water molecule
- Q. 5. Explain the solubility with reference to "like dissolve like" principle.

Section D

- Q. 1. What is the molarity of the solution prepared by dissolving 1.25 g of HCl gas into enough water to make 30 cm³ of solution?
- Q. 2. A solution of potassium chloride was prepared by dissolving 2.5 g of potassium chloride (KCl) in water and making the volume up to 100

cm³. Find the concentration of solution in mol/ dm³.

Q. 3. A flask contains 0.25 M NaOH solution. What mass of NaOH is present per dm³ of solution?

Q. 4. What volume of 0.5M acid is needed to neutralize 200ml of 4M base?

Q. 5. A mineral water bottle contains 28 mg of calcium in 100 cm³ of solution. What is the concentration in g/dm³?

Q. 6. A solution of 20cm³ of alcohol is dissolved in 80cm³ of water. Calculate the concentration (v/v) of this solution.

Q. 7. How much sodium hydroxide (NaOH) is required to prepare 400 cm³ of 0.3M solution?

Chapter 7: ELECTROCHEMISTRY

Section B

12. What is redox reaction? Give an example of it with equation and also identify the oxidizing and reducing agents in the equation.	2025
5. Define oxidizing and reducing agents. Identify oxidizing and reducing agents of the following chemical reaction. $\text{Zn} + \text{Cl}_2 \longrightarrow \text{ZnCl}_2$	2023
10. State Faraday's 1, law of electrolysis and explain it	2022
14. A current of 5 Ampere was passed through an electrolytic solution of Copper Sulphate for an hour. Find the mass of Copper metal deposited at the Cathode. (The Electro chemical Equivalent of Copper is 0.000329g/c)	2018
9. State Faraday's 1 st law of electrolysis and 2 nd law of Electrolysis. 10. Write three differences between Oxidation and Reduction. 14. Define Electrolysis and write its two uses.	2017
19. What is an Alloy? Write names of two alloys of Copper with their composition.	2016
16. Define Oxidation by three different ways with one suitable example for each. 20. Define the following: (i) Electrochemical Equivalent (ii) Faraday	2015

Section C

16. Describe the construction of a dry cell with diagram. Explain its working with the help of chemical equations.	2025
15. Define electroplating. Describe the Tin plating of an Iron spoon with the help of ionic equations. Draw its labelled diagram	2024

also.

14. Explain the corrosion of Iron and describe any four methods to prevent metals from it **2023**

16. What is electroplating? How can a steel spoon be electroplated with chromium explain with electrodes reaction also draw a labelled diagram of electrolytic cell. **2022**

14. Explain with illustrations what are Electrolyte and non-Electrolyte and how they are classified? **2021**

20. (b) State Faraday's First and Second law of Electrolysis. Calculate the amount of Silver deposited at Cathode when 10 ampere of current is passed for 50 minutes through the solution of silver Nitrate. **2019**

20. (b) What is electroplating? Explain how would you electroplate an Iron spoon with Nickel? Draw a labelled diagram also. **2018**

25 (c). Calculate the amount of Silver (Ag) Deposited at Cathode, When 10 Ampere of Electric current is passed for 1 hour through the solution of $AgNO_3$. (Z of Ag=0.001118 g/C) **2016**

SOME IMPORTANT QUESTIONS FROM NEW BOOK

Section B

Q. 1. Define Oxidation, reduction reaction with examples.

Q. 2. Why ionic compound conduct electricity in molten or in aqueous solutions only?

Q. 3. What is electrolytic cell? Explain with diagram.

Q. 4. Define oxidizing and reducing agent with examples.

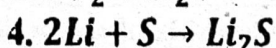
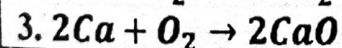
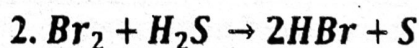
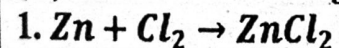
Q. 5. Examine the following chemical equation and identify.

(i) Oxidizing agent

(ii) Reducing agent

(iii) Substance undergoes oxidation.

(iv) Substance Undergoes Reduction.



Q. 6. Identify the Alloy.

COMPONENTS	ALLOY
Cu-Zn	
Cu - Al -Mg -Ni	
Cu - Zn -Sn	

Section C

Q. 1. Describe the dry cell with diagram.

Q. 2. What is Battery? How lead storage Battery works?

Q. 3. Explain process of electrolysis in electrolytic cell.

Q. 4. What is Alloy? Explain its classification with examples.

Q. 5. What is rusting? How it can be prevented.

Q. 6. What is Electroplating? How steel object can be electroplated with Tin, Zinc and Silver?

Q. 7. State and explain Faraday first and second law of electrolysis.

Chapter 8: CHEMICAL REACTIVITY

Section B

22. Define: (ii) Halogens

2017

18. Write three characteristics of Halogens.

2016

Section C

SOME IMPORTANT QUESTIONS FROM NEW BOOK

Section B

Q. 1. Identify the elements as Metals, Non-metals and Metalloids from the following elements:-

ELEMENTS	METALS	NON-METALS	METALLOIDS
C, Ca, Sb, S, Sr, Si, K, P, Ba, Ge			

Q. 2. What are Alkali metals? Write names and symbols of alkali metals?

Q. 3. What are Alkaline Earth Metals? Write names and symbols of Alkaline earth metals?

Q. 4. What are Halogens? Write names and symbols of Halogens?

Q. 5. Write names and symbols of any three metalloids.

Q. 6. Define following with examples: - 1. Cation 2. Anion

Q. 7. Explain why alkali metals do not exist free in nature?

Q. 8. Explain inertness of noble metals.

Q. 9. Write names and symbols of some noble metals

Q. 10. Explain electronegative character of non-metals.

Section C

Q. 1. Explain importance of Silver.

Q. 2. Explain importance of Gold.

Q. 3. Explain the experiment to test Cl and I Ions.

Q. 4. Explain Electropositive character of metals.

Q. 5. Explain the position of Magnesium in periodic table and its importance.

Q. 6. Explain the position of Sodium in periodic table and its importance.

Q. 7. Arrange the following Halogen Acids in increasing order of their

Acidic strength: HBr, HCl, HI, HF.

Q. 8. Explain Electronegative character of non-metals.

Q. 9. Differentiate between sodium and iron as soft and hard metal.

Q. 10. Discuss the reactivity of Halogens.

Section D

DEFINITIONS

QUESTIONS	CH	Years
9. Define the following terms: (i) Free radical	1	2025
(ii) Solubility	6	
(iii) Alloy.	8	
2. Define the following terms:	3	2024
(i) Periodicity (ii) Electronegativity		
(iii) Electrolysis	7	
2. Define: (i) Tyndall effect	6	2022
(ii) Valency	1	
(iii) Electronegativity	3	

SCIENTIFIC REASONS

QUESTIONS	CH	Years
13. Give reasons:	2	2025
(i) How does electron differ from neutron?		
(ii) Why the rate of diffusion of gases is rapid than that of liquids?	5	
13. Justify the following:	3	2024
(i) Why Fluoride ion is not considered as Neon atom?		
(ii) Why noble gases do not react with other elements at ordinary conditions.		
13. Justify the followings:	4	2023
(i) Most of Ionic compounds are found in solid states.		
(ii) Alkali metals form cation easily.		
13. Give reason:	6	2022
(i) Why gasoline does not dissolve in water?		
(ii) Alkali metals are not found in Free State in nature, why?	3	
15. Give reason of the following:	3	2019
(i) Why does ionization energy of elements increase from left to right in a period of the periodic table		